

Customer Shopping Behavior Analysis

1.Project Overview

This project analyzes customer shopping behavior using transactional data from 3900 purchases across various products categories .The goal is to uncover insights into spending patterns, customer segments, product preferences and subscription behavior to guide strategic business decisions.

2.Dataset Summary

Rows: 3900

Columns:18

Key features:

- Customer demographics(Age, Gender, Location, Subscription status)
- Purchase details(Item purchased, category, purchase amount, season,size,color)
- Shopping behavior(Discount applied, promo code used, previous purchases, frequency of purchases, Review rating, Shipping type)
- Missing Data: 37 values in Review Rating column.

3.Exploratory Data Analysis using Python

I began with data preparation and cleaning in Python:

- Data loading: Imported the dataset using **pandas**.
- Initial Exploration: Used **df.info()** to check structure and **df.describe()** for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6	7
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

- Database Integration: connected Python script to MySQL and loaded the cleaned Data frame into the database for SQL analysis.

4.Data analysis using SQL

1.Revenue by Gender-compared total revenue generated by male vs. female customers.

	Gender	Revenue
▶	Male	155438
	Female	75191

2. High spending discount users-identified customers who used discounts but still spend above the average purchase amount.

	Gender	Purchase_Amount
▶	Male	64
	Male	73
	Male	90
	Male	85
	Male	97
	Male	68
	Male	72
	Male	81
	Male	90
	Male	62

3.Which are the top 5 products with the highest average review rating?

	Item_Purchased	Average_Rating
▶	Gloves	3.8627737226277383
	Sandals	3.8446540880503144
	Boots	3.818881118881119
	Hat	3.801307189542483
	Skirt	3.7853503184713366

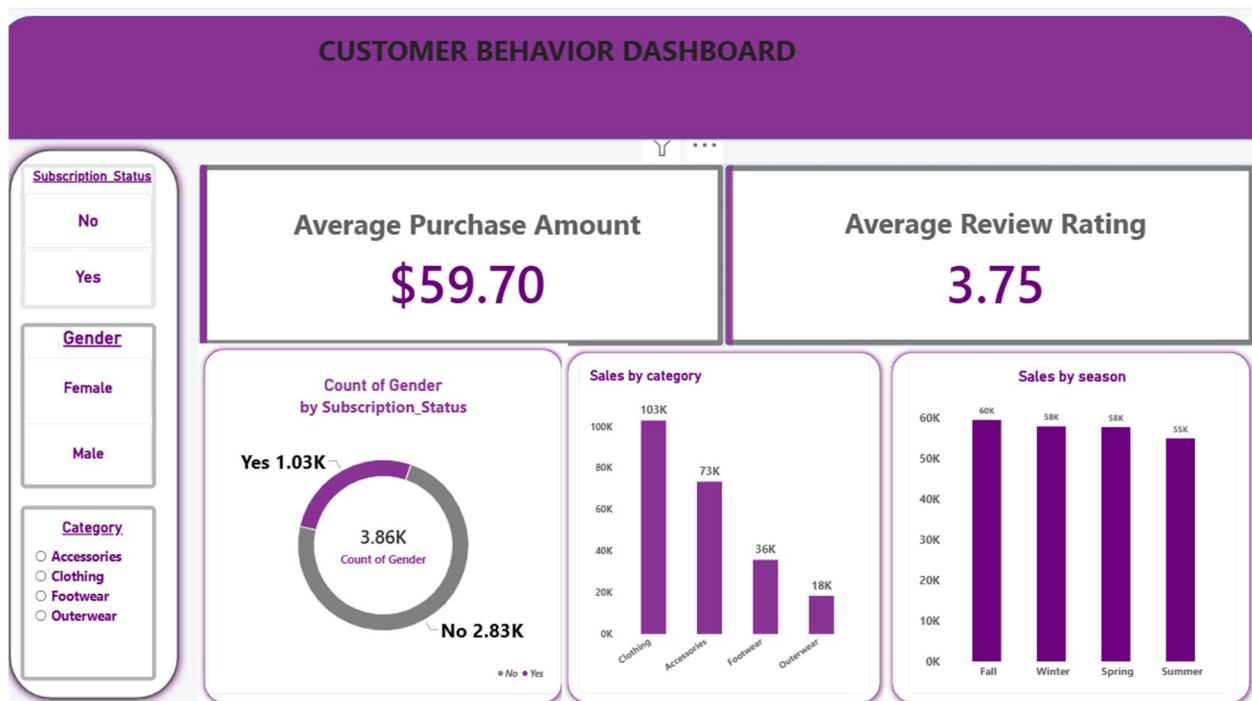
4.Comparing the average purchase amount between standard and express shipping.

	Shipping_Type	Average_purchase_amount
▶	Express	60.3443
	Standard	58.4138

5. Compared whether subscribed customers spend more. I compared average amount spend and total revenue between subscribers and non-subscribers.

	Subscription_Status	customers	Revenue	Average_spent
►	Yes	1033	61377	59.42
	No	2830	169252	59.81

6. Data analysis using PowerBI and preparation of dashboard.



7. Business Recommendations

- Boost subscriptions-promote exclusive benefits for subscribers.
- Customer loyalty programs-reward repeat buyers.
- Target marketing-focus efforts on high revenue customers and express shipping.